

ROSLAVL, Russian Federation

WMO: 268820

Lat: 53.933N

Lon: 32.833E

Elev: 736

StdP: 14.31

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-1.7	-12.3	2.9	-6.4	-6.5	4.0	-0.7	23.8	30.8	21.0	30.0	5.1	110	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	83.7	67.8	80.4	66.5	77.4	65.0	69.8	80.2	68.1	77.5	66.5	75.1	6.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	99.1	74.6	64.6	93.9	72.6	63.2	89.2	70.9	34.1	80.4	32.8	77.6	31.5	75.3	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.0	15.0	-13.4	87.6	5.1	4.4	-17.0	90.8	-20.0	93.4	-22.9	95.8	-26.6	99.0		
(5)			-13.7	72.3	5.1	1.7	-17.3	73.5	-20.3	74.5	-23.1	75.4	-26.8	76.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	20.6	21.7	30.7	44.8	55.4	61.7	66.2	63.0	53.3	42.8	32.6	24.2		
	DBStd	18.17	10.92	11.13	8.28	8.39	8.38	6.32	5.69	6.24	6.57	7.35	8.88	10.64		
	HDD50	4152	911	793	599	201	41	1	0	0	40	244	522	800		
	HDD65	8158	1376	1213	1064	607	314	138	54	113	354	688	972	1265		
	CDD50	1669	0	0	1	45	208	351	501	404	139	20	0	0		
	CDD65	199	0	0	0	0	16	38	90	52	3	0	0	0		
	CDH74	1552	0	0	0	5	130	267	680	443	27	0	0	0		
	CDH80	350	0	0	0	1	18	36	165	128	2	0	0	0		
	WSAvg	7.1	8.3	8.2	8.3	7.1	6.7	6.5	5.4	5.8	5.9	7.2	7.9	8.2		
	PrecAvg	23.8	1.3	0.9	1.2	1.4	1.9	3.4	3.3	2.4	2.3	2.1	1.9	1.5		
(15) Precipitation	PrecMax	31.2	3.0	2.3	2.9	3.5	5.0	5.9	7.0	7.2	7.5	4.8	3.8	3.2		
	PrecMin	16.1	0.1	0.1	0.0	0.2	0.1	0.6	0.6	0.4	0.2	0.0	0.0	0.4		
	PrecStd	4.0	0.7	0.5	0.7	0.8	0.9	1.5	1.6	1.7	1.5	1.2	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	40.8	57.6	72.6	82.7	83.6	88.8	91.0	77.6	67.1	53.5	46.6		
		MCWB	39.1	37.1	46.1	54.3	66.4	66.4	70.1	69.0	62.8	54.6	50.6	45.4		
	2%	DB	37.2	37.6	50.1	67.5	77.9	80.4	84.3	82.8	71.9	60.5	48.6	42.1		
		MCWB	35.7	35.6	41.3	52.4	62.9	66.3	68.4	68.3	60.2	55.3	46.7	40.9		
	5%	DB	34.9	35.6	45.4	63.4	73.5	77.0	80.9	78.6	67.7	56.2	45.8	38.1		
		MCWB	33.8	33.8	39.1	50.7	61.0	64.4	67.2	65.5	58.5	51.9	44.3	37.0		
	10%	DB	33.2	34.2	41.3	59.0	69.8	73.8	77.5	74.8	64.1	53.2	43.7	35.6		
		MCWB	32.3	32.9	36.6	48.6	57.8	63.3	66.0	63.7	56.5	49.9	42.3	34.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	38.6	46.8	56.5	67.4	69.7	72.5	71.7	65.7	59.2	51.5	45.8		
		MCDB	39.8	40.2	55.3	67.6	79.2	79.4	84.2	82.6	75.5	63.4	52.8	46.3		
	2%	WB	35.9	36.0	42.8	53.8	64.5	67.8	70.3	69.3	62.0	55.5	47.1	40.9		
		MCDB	36.8	37.0	48.2	63.8	75.0	76.9	80.9	79.9	69.8	59.2	48.0	41.8		
	5%	WB	34.0	34.3	39.8	52.0	62.2	66.2	68.6	67.4	59.6	52.7	44.7	37.3		
		MCDB	34.7	35.2	44.4	61.0	71.2	74.8	77.9	77.1	65.0	55.8	45.7	37.9		
	10%	WB	32.4	33.2	37.1	49.8	59.9	64.3	67.0	65.0	57.4	50.4	42.3	34.7		
		MCDB	33.1	34.1	40.7	57.9	67.8	72.0	75.1	72.4	62.5	52.8	43.5	35.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.7	11.8	15.8	17.3	16.2	16.8	17.2	15.6	11.5	7.3	7.2		
		MCDBR	6.3	8.0	17.2	21.9	22.3	20.2	21.5	22.6	21.4	16.1	9.0	7.0		
		MCWBR	6.3	6.9	10.4	10.6	10.5	9.2	8.7	9.7	11.2	10.6	7.6	6.6		
	5% WB	MCDBR	5.9	7.1	15.3	19.0	20.4	18.3	19.2	20.6	17.9	14.7	7.7	7.0		
		MCWBR	6.1	6.6	10.1	9.9	10.0	9.3	8.7	9.7	10.6	10.7	7.2	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.324	0.349	0.415	0.393	0.377	0.408	0.421	0.376	0.350	0.328	0.292		
	taud		2.132	2.072	2.125	2.144	2.303	2.366	2.299	2.252	2.348	2.380	2.255	2.152		
	Ebn at Noon		204	235	259	256	271	277	264	252	250	230	190	177		
	Edh at Noon		21	32	37	43	38	36	38	38	30	24	19	17		
(44) All-Sky Solar Radiation	RadAvg		223	511	964	1273	1710	1809	1741	1439	958	464	175	132		
	RadStd		47	83	131	132	116	161	153	159	95	94	36	36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+1.15	N/A	N/A	N/A	N/A	N/A	-335	+166	N/A

Nomenclature: See separate page